Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Published IEEE author with research experience in Continual Learning and Vision Transformers. Hands-on experience developing machine learning, computer vision, and Generative AI applications using Python, PyTorch, FastAPI, Docker, Linux, and Git. Experienced in building end-to-end AI pipelines, model evaluation frameworks, and AI-powered document intelligence solutions. Seeking Data Scientist, AI Engineer, or Machine Learning Engineer opportunities.

EDUCATION

Sapienza Università di Roma **May 2026**

Master's in Computer Science

- **Relevant Coursework:** Computer Vision, Deep Learning and Applied AI, Natural Language Processing, Machine Learning, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

SKILLS

- **Programming :** Python, C++, SQL, Bash
- **Computer Vision :** Image Processing, Feature Extraction, OpenCV, CNNs, Vision Transformers, Image Classification, Visual Inspection, Defect Detection
- **Machine Learning :** Deep Learning, Model Evaluation, Feature Engineering, Transfer Learning, Continual Learning, Transformer Architectures
- **Frameworks & Libraries :** PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn
- **Research & Mathematics :** Experimental Design, Scientific Literature Review, Performance Analysis, Scientific Writing, Linear Algebra, Optimization Fundamentals
- **Tools:** Git, GitHub, Linux, Docker, AWS

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) Nov 2025

- Evaluated pretrained Vision Transformer architectures under Task-Incremental, Domain-Incremental, and Class-Incremental learning settings.
- Designed experimental pipelines for training, evaluation, and comparison of computer vision models across sequential tasks.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.
- Conducted extensive review and analysis of state-of-the-art research literature in continual learning and computer vision.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.

Machine Learning for Air Pollution [IEEE Publication] May 2023

- Published IEEE conference paper evaluating machine learning approaches for environmental prediction and monitoring.
- Analyzed Random Forest, SVM, Genetic Algorithm, LS-SVM, Decision Trees, and Artificial Neural Networks across multiple air quality datasets.
- Performed comparative evaluation of model performance and identified opportunities for future research and real-time monitoring applications.
- Presented scientific findings through structured research documentation and publication.

PROJECTS

Metal Defect Detection using Computer Vision

- Developed and evaluated computer vision models for automated metal surface defect detection using the NEU Metal Surface Defects dataset.
- Designed a comparative study of CNN, Support Vector Machine (SVM), and Random Forest approaches to analyze the effectiveness of deep learning versus traditional machine learning techniques for industrial inspection.
- Implemented image preprocessing, data augmentation, feature extraction, and model training pipelines using TensorFlow, Keras, NumPy, and Scikit-learn.
- Performed systematic model evaluation using accuracy metrics, confusion matrices, and class-level performance analysis across multiple defect categories.
- Investigated the applicability of computer vision techniques for automated visual inspection, defect classification, and quality-control systems in manufacturing environments.
- Investigated computer vision approaches relevant to defect classification and object detection tasks in industrial inspection settings.

DocDelta [Document Comparison API]

- Developed an NLP system using Sentence Transformers for semantic comparison and change detection in legal documents.
- Designed evaluation pipelines to identify added, removed, and modified content using embedding similarity techniques.
- Deployed solution using FastAPI, Docker, and AWS ECS Fargate.
- Implemented monitoring and performance analysis using AWS CloudWatch.

Lossy Text Compressor [Semantic NLP Pipeline]

- Investigated semantic text compression using transformer embeddings, quantization, and reconstruction techniques.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics.
- Analyzed trade-offs between compression efficiency and semantic preservation across multiple quantization levels.

RESEARCH INTERESTS

Computer Vision, Visual Intelligence, Image Processing, Vision Transformers, Continual Learning, Automated Defect Detection, Deep Learning, Applied AI Research